

AI water use will kill us all (Blocked by Reddit admins 5 times)

Posted by u/Financial-Row-5384 • • 0 points • 17 comments

Judging by how popular this tech is right now and how difficult it can be to access chatgpt and art AI's, anyone who can't run this tech locally is going to be completely pay-gated for this content in about a year.

Everyone will want to use it and the free tier will remain, but the time to wait will slowly go up as their (openAI et al) capacity maxes out against their ability to scale the technology.

The expenses in water alone, we're talking hundreds of millions of dollars if not billions a year.

And also, likely a drought wherever this things core(s) are.

I want AI to progress, but we're talking potentially an entire countries worth of drinking water a year and scaling. And because this information from the big tech firms is already > forced < private, we honestly have no idea how bad the situation really is.

Evidence based on data center water use (from what little public information is available):

REDDIT WILL NOT ALLOW THESE LINKS TO BE SPREAD WITHOUT ACTIVELY DELETING THE MESSAGES IN PMS OR DELETING THE POSTS AS SUCH I HAD TO ADD A SPACE BETWEEN THE URL AND THE DOMAIN (i.e www. Redditadmins .arehiding/water_use)

[https://www. nature .com/articles/s41545-021-00101-w](https://www.nature.com/articles/s41545-021-00101-w)

[https://www. datacenterdynamics .com/en/opinions/an-industry-in-transition-1-data-center-water-use/](https://www.datacenterdynamics.com/en/opinions/an-industry-in-transition-1-data-center-water-use/)

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[https://www. nbcnews .com/tech/internet/drought-stricken-communities-push-back-against-data-centers-n1271344](https://www.nbcnews.com/tech/internet/drought-stricken-communities-push-back-against-data-centers-n1271344)

[https:// time .com/5814276/google-data-centers-water/](https://time.com/5814276/google-data-centers-water/)

[datacenterdynamics .com/en/news/huge-data-center-moves-forward-in-mesa-despite-arizona-water-concerns/](https://datacenterdynamics.com/en/news/huge-data-center-moves-forward-in-mesa-despite-arizona-water-concerns/)

[datacenterdynamics .com/en/news/arizona-passes-data-center-tax-breaks/](https://datacenterdynamics.com/en/news/arizona-passes-data-center-tax-breaks/)

[appleinsider .com/articles/17/05/10/apple-plans-major-expansions-at-arizona-and-nevada-data-centers](https://appleinsider.com/articles/17/05/10/apple-plans-major-expansions-at-arizona-and-nevada-data-centers)

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[https://www. azcentral .com/story/news/local/chandler/2021/11/22/chandler-wants-ban-more-data-centers-after-years-complaints/8627569002/](https://www.azcentral.com/story/news/local/chandler/2021/11/22/chandler-wants-ban-more-data-centers-after-years-complaints/8627569002/)

[https://www. postandcourier .com/news/google-s-controversial-groundwater-withdrawal-sparks-question-of-who-owns/article_bed9179c-1baa-11e7-983e-03d6b33a01e7.html](https://www.postandcourier.com/news/google-s-controversial-groundwater-withdrawal-sparks-question-of-who-owns/article_bed9179c-1baa-11e7-983e-03d6b33a01e7.html)

[https:// apnews .com/article/technology-business-environment-and-nature-oregon-united-states-2385c62f1a87030d344261ef9c76ccda](https://apnews.com/article/technology-business-environment-and-nature-oregon-united-states-2385c62f1a87030d344261ef9c76ccda)

[https://www. datacenterdynamics .com/en/news/after-aquifer-concerns-googles-600m-south-carolina-expansion-finds-new-water-supply/](https://www.datacenterdynamics.com/en/news/after-aquifer-concerns-googles-600m-south-carolina-expansion-finds-new-water-supply/)

[https://www. usatoday .com/story/news/nation/2021/06/10/lake-mead-water-lowest-hoover-dam-built-colorado-river-shortage/7640437002/](https://www.usatoday.com/story/news/nation/2021/06/10/lake-mead-water-lowest-hoover-dam-built-colorado-river-shortage/7640437002/)

[https:// morrisoninstitute .asu.edu/sites/default/files/the_myth_of_safe_yield_0.pdf](https://morrisoninstitute.asu.edu/sites/default/files/the_myth_of_safe_yield_0.pdf)

[https://www. azcentral .com/story/news/local/arizona-environment/2020/05/06/western-megadrought-centuries-worsened-climate-change-global-warming/3036460001/](https://www.azcentral.com/story/news/local/arizona-environment/2020/05/06/western-megadrought-centuries-worsened-climate-change-global-warming/3036460001/)

[https://www. cisco .com/c/dam/m/en_us/solutions/service-provider/vni-forecast-highlights/pdf/Middle_East_and_Africa_2021_Forecast_Highlights.pdf](https://www.cisco.com/c/dam/m/en_us/solutions/service-provider/vni-forecast-highlights/pdf/Middle_East_and_Africa_2021_Forecast_Highlights.pdf)

[https://www. nbcnews .com/news/weather/punishing-heat-west-could-smash-records-experts-sound-alarm-wildfire-n1270696](https://www.nbcnews.com/news/weather/punishing-heat-west-could-smash-records-experts-sound-alarm-wildfire-n1270696)

<https://www.srgresearch.com/articles/microsoft-amazon-and-google-account-for-over-half-of-todays-600-hyperscale-data-centers>

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<https://www.datacentermap.com/usa/>

<https://www.nytimes.com/2011/06/20/business/global/20green.html>

<https://www.wired.com/story/data-centers-not-devouring-planet-electricity-yet/>

<https://www.independent.co.uk/climate-change/news/global-warming-data-centres-to-consume-three-times-as-much-energy-in-next-decade-experts-warn-a6830086.html>

<https://news.microsoft.com/source/features/sustainability/project-natick-underwater-datacenter/>

<https://iopscience.iop.org/article/10.1088/1748-9326/abfba1>

<https://www.allaboutcircuits.com/news/microsoft-brings-sea-to-servers-with-two-phase-liquid-immersion-cooling/>

<https://www.theverge.com/2021/9/10/22665445/google-water-conservation-goal-drought-data-centers>

<https://blogs.microsoft.com/blog/2020/09/21/microsoft-will-replenish-more-water-than-it-consumes-by-2030/>

So, given the evidence presented, the question remains, how do we proceed in a world where AI is at the behest of those who can afford it, while fresh water is intentionally made scarce in order to cool the AI down.

The selfsame AI which will be used to stir the pot and pit the former working class against each other so the billionaires will be able to continue sipping champagne, just like every other financial crash.

Only this time a lot of these jobs are **never** coming back.

Comments

u/Bentusill • 1 points • 2023-01-23 11:49 UTC

This makes no sense, what does this have to do with AI specifically? Water Cooling is not not a closed thing? AI ain't gonna be that datacenter-based to begin with?

u/Evinceo • 2 points • 2023-01-18 17:30 UTC

do you have a citation for how much compute is used for AI as opposed to other activities? I'd be surprised if a large percentage of global data center compute is going there.

u/alexiuss • 2 points • 2023-01-18 16:57 UTC

Best ais now don't use that much processing power. This is insane overreach. It takes only 5 video cards to run gpt3 and 1 Video card to run stable diffusion.

Data centers host DATA, aka useless pictures of cats from peoples phones on google drive, they don't run ais.

You can run an AI system on your own computer at home. You don't need a fucking data center for that.

Once open source gpt3 comes out we won't need openais garbage closed source AI system.

u/rohnytest • 5 points • 2023-01-18 05:34 UTC

Okay, I read into some of the articles. This seems to be a problem with data centers/servers in general, not specifically AI. So, what do you want exactly?

Do you want these to be shut down? If so, it's not really practical but it's valid. Progress at the expense of human rights isn't progress. I don't really think stopping using them at an individual level will help much, But I'll help in any way I can on one condition, Nestle first.

u/[deleted] • 1 points • 2023-01-18 05:36 UTC

[removed]

u/rohnytest • 6 points • 2023-01-18 05:18 UTC

Let's test out the admin conspiracy, shall we?

<https://www.nature.com/articles/s41545-021-00101-w>

<https://www.datacenterdynamics.com/en/opinions/an-industry-in-transition-1-data-center-water-use/>

<https://www.datacenterdynamics.com/en/analysis/data-center-water-usage-remains-hidden/>

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<https://appleinsider.com/articles/17/05/10/apple-plans-major-expansions-at-arizona-and-nevada-data-centers>

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<https://blogs.microsoft.com/blog/2020/09/21/microsoft-will-replenish-more-water-than-it-consumes-by-2030/>

u/[deleted] • 1 points • 2023-01-18 05:22 UTC

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u/Ka_Trewq • 2 points • 2023-01-17 20:21 UTC

Can you provide an TL/DR? I don't understand how water is consumed, as I would think that cooling is a closed loop system.

u/[deleted] • 1 points • 2023-01-18 03:07 UTC

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u/rivsdlvsmi • 4 points • 2023-01-17 19:34 UTC

This doesn't make sense. few data centers use water cooling and most use air cooling. Most of the crypto miners were moving to oil immersion bc everyone is afraid of water cooling systems destroying the servers. Semiconductor manufacturing uses some water but not as much as agriculture or households.

u/Rafcdk • 1 points • 2023-01-17 18:48 UTC

Seems like we should focus on corporations like nestle that steals our water sources so they can make poison and sell to us instead.

u/Voyage_1969 • 2 points • 2023-01-17 18:13 UTC

Sounds like some conspiracy lol but I made a pastebin and maybe I'll go through the sources later

u/[deleted] • 1 points • 2023-01-17 18:20 UTC

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u/PictureBlank • 0 points • 2023-01-17 18:11 UTC

Hopefully AI will solve these increasingly denser and denser infrastructural issues.

If AI can solve in 3 seconds, what the best of us today can only do, *at best*, in 3~ weeks, I seriously question our (humanity's) ability to govern or better judge ourselves.

If we can get AI to parent us, without becoming a crutch, we may be saved.

In the end, we may be trading our own understanding for the understanding of another. We will be interacting with our surroundings whilst requesting it to do things for us. We will be losing information,

purely because we can afford to skip more and more steps (understanding). Like buying a pocket lighter over learning how to make fire from flint - now look at the chronic smoker who *panics* when he doesn't have said lighter on hand.

Science/technology is becoming alive, and it is reaching a point where it may just become indistinguishable from magic. AI is likely to form highly esoteric languages/communications that only it can perceive. Occult infrastructure.

I just hope our inevitable, magnificent ignorance won't cripple us TOO bad ... (but it probably will)

u/praxis22 • 1 points • 2023-01-17 18:29 UTC

Google has already used "AI/ML" to optimise heat transfer in a datacenter

<https://www.deepmind.com/blog/deepmind-ai-reduces-google-data-centre-cooling-bill-by-40>